

It is illustrated in Fig. 20, which shows several pairs of graphs of the initial and inverse functions. A list of ^{some} pairs of functions with their domains is given below:

(a) initial $y = x^3, -\infty < x < \infty$
 inverse $y = \sqrt[3]{x}, -\infty < x < \infty$

(b) initial $y = x^2, 0 \leq x < \infty$
 inverse $y = \sqrt{x}, 0 \leq x < \infty$

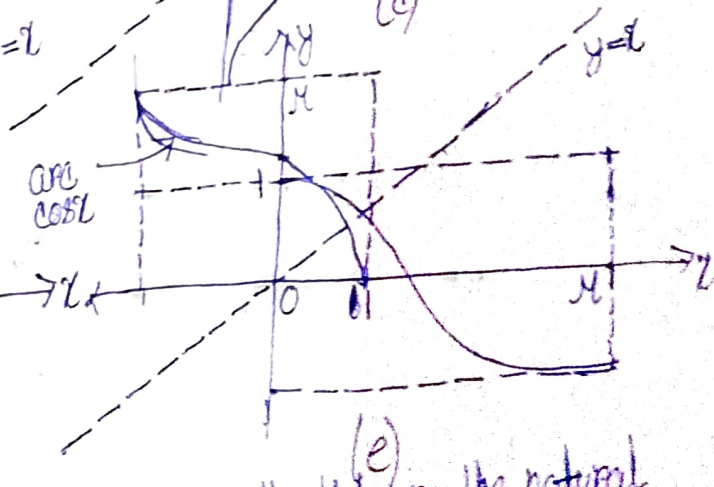
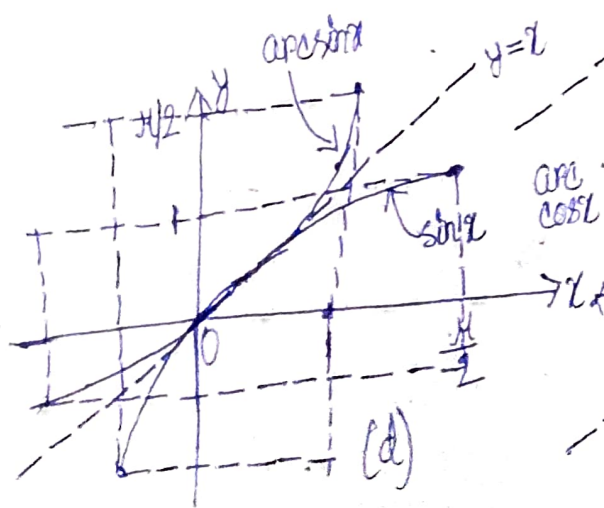
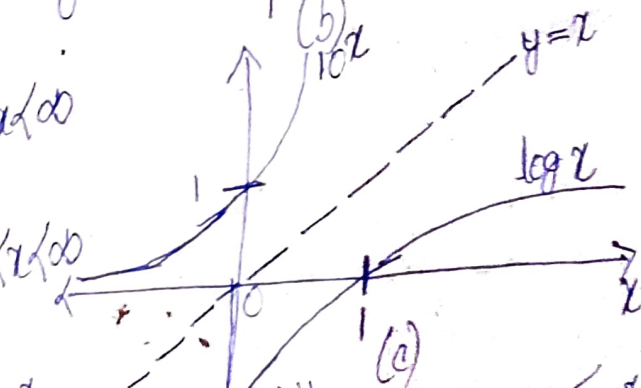
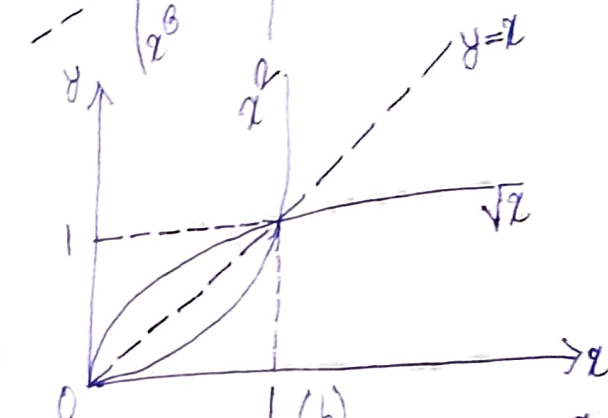
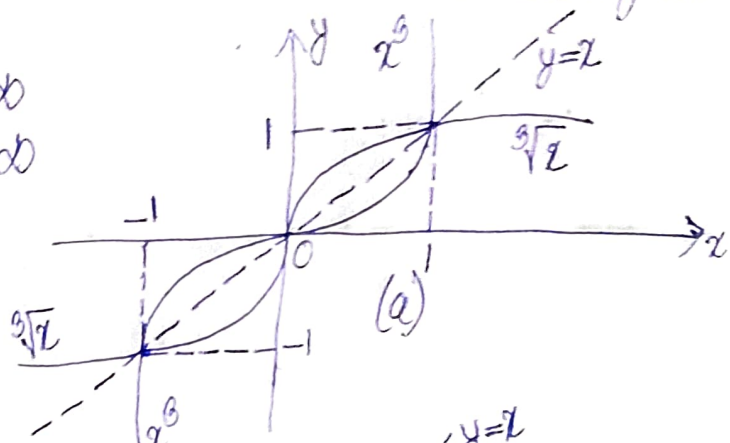
(c) initial $y = 10^x, -\infty < x < \infty$
 inverse $y = \log x, 0 < x < \infty$

(d) initial $y = \sin x, -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$
 inverse $y = \arcsin x, -1 \leq x \leq 1$

(e) initial $y = \cos x, 0 \leq x \leq \pi$
 inverse $y = \arccos x, -1 \leq x \leq 1$

(f) initial $y = \tan x, -\frac{\pi}{2} < x < \frac{\pi}{2}$
 inverse $y = \arctan x, -\infty < x < \infty$

(g) initial $y = \cot x, 0 < x < \pi$
 inverse $y = \operatorname{arccot} x, -\infty < x < \infty$



All the domains of the inverse functions shown in the list are the natural domains of the functions (however, in the case of $y = \sqrt[3]{x}$ the natural domain is sometimes assumed to be restricted to the interval $[c, \infty[$ instead of the whole real line). As to the initial functions, only two of them ($y = x^2$ and $y = 10^x$) are considered in this case as defined on their natural domains.